

ANOVA: Two-Factor With Replication

SS Within — Easy, Line-by-Line Calculation

Goal: Calculate SS Within using the easy method: Mean → Subtract → Square → Add.

Method	Morning	Evening
A	72, 75, 74	68, 70, 69
B	80, 82, 81	76, 78, 77
C	88, 90, 89	84, 86, 85

What is SS Within?

SS Within measures how far the individual values are from the average of their own small group (cell).

Easy rule for every cell: ① Find mean → ② Subtract mean → ③ Square → ④ Add.

Formula: **SS for one cell** = $\Sigma(X - \text{Cell Mean})^2$

Finally: **SS Within** = all 6 cell SS values added together.

1. Method A + Morning

Values: **72, 75, 74** Cell Mean: **73.6667**

Mean = $(72 + 75 + 74) / 3 = 221 / 3 = 73.6667$

$72 - 73.6667 = -1.6667 \rightarrow \text{square} = 2.7778$

$75 - 73.6667 = 1.3333 \rightarrow \text{square} = 1.7778$

$74 - 73.6667 = 0.3333 \rightarrow \text{square} = 0.1111$

SS = $2.7778 + 1.7778 + 0.1111 = 4.6667$

2. Method A + Evening

Values: **68, 70, 69** Cell Mean: **69**

Mean = $(68 + 70 + 69) / 3 = 207 / 3 = 69$

$68 - 69 = -1 \rightarrow \text{square} = 1$

$70 - 69 = 1 \rightarrow \text{square} = 1$

$69 - 69 = 0 \rightarrow \text{square} = 0$

SS = $1 + 1 + 0 = 2$

3. Method B + Morning

Values: **80, 82, 81** Cell Mean: **81**

Mean = $(80 + 82 + 81) / 3 = 243 / 3 = 81$

$80 - 81 = -1 \rightarrow \text{square} = 1$

$$82 - 81 = 1 \rightarrow \text{square} = 1$$

$$81 - 81 = 0 \rightarrow \text{square} = 0$$

$$SS = 1 + 1 + 0 = 2$$

4. Method B + Evening

Values: **76, 78, 77** Cell Mean: **77**

$$\text{Mean} = (76 + 78 + 77) / 3 = 231 / 3 = 77$$

$$76 - 77 = -1 \rightarrow \text{square} = 1$$

$$78 - 77 = 1 \rightarrow \text{square} = 1$$

$$77 - 77 = 0 \rightarrow \text{square} = 0$$

$$SS = 1 + 1 + 0 = 2$$

5. Method C + Morning

Values: **88, 90, 89** Cell Mean: **89**

$$\text{Mean} = (88 + 90 + 89) / 3 = 267 / 3 = 89$$

$$88 - 89 = -1 \rightarrow \text{square} = 1$$

$$90 - 89 = 1 \rightarrow \text{square} = 1$$

$$89 - 89 = 0 \rightarrow \text{square} = 0$$

$$SS = 1 + 1 + 0 = 2$$

6. Method C + Evening

Values: **84, 86, 85** Cell Mean: **85**

$$\text{Mean} = (84 + 86 + 85) / 3 = 255 / 3 = 85$$

$$84 - 85 = -1 \rightarrow \text{square} = 1$$

$$86 - 85 = 1 \rightarrow \text{square} = 1$$

$$85 - 85 = 0 \rightarrow \text{square} = 0$$

$$SS = 1 + 1 + 0 = 2$$

7. Add All Six Cell SS Values

Now add the six answers:

$$4.6667 + 2 + 2 + 2 + 2 + 2 = \mathbf{14.6667}$$

Cell	SS
A + Morning	4.6667
A + Evening	2
B + Morning	2
B + Evening	2
C + Morning	2
C + Evening	2
SS Within	14.6667

Easy Method to Remember

For every small group: **Mean** → **Subtract** → **Square** → **Add**.

Repeat for all 6 cells, then add the six SS values.

Important: SS Within uses the **cell mean**, not the Grand Mean.

Final Answer: SS Within = 14.6667